

Peer Reproducibility Check

ExInt II: Research Designs in SME Research · WU Vienna · SS 2026

Voluntary — strongly recommended before June 28, 2026

Reference project:

<https://github.com/vkiefner/sme-intl/tree/main>

Where to work:

Terminal

VS Code

Browser

GitHub

This checklist walks you through cloning a classmate's project, running their full pipeline from scratch, and verifying that their research note compiles to a PDF. This is exactly what the examiner will do. Work in pairs. Use the same operating system as your partner (Windows with Windows, Mac with Mac). You will need your own WRDS credentials — the .env file is never committed to Git.

A Before you start — what you need

Browser

GitHub

Step 1 — Exchange GitHub repo URLs with your partner

Ask your partner for their GitHub repo URL. It should look like:

```
https://github.com/[their-username]/[their-repo-name]
```

Open it in your browser and confirm the repo is public and contains code/, research_note.qmd, and Taskfile.yml.

■ If the repo is private, the examiner cannot access it. Tell your partner to make it public: **GitHub** → **Settings** → **Danger Zone** → **Change visibility**.

Terminal

Step 2 — Choose a working directory — NOT inside your own project

Clone into a separate folder so it does not interfere with your own project:

```
cd C:\Users\[you]\dev\python\peer-check
```

■ Mac / Linux:

```
cd ~/dev/peer-check
```

■ Create the folder first if it does not exist: `mkdir peer-check`

Terminal

Step 3 — Confirm you have WRDS credentials

You will need to add your own .env file after cloning. Confirm you know your WRDS username:

```
# Your WRDS username (the one you use to log in to  
wrds-www.wharton.upenn.edu)
```

```
# You will need this in Step 8
```

■ The .env file is intentionally excluded from Git. You must create it yourself after cloning.

B Clone the repo and set up the environment

Terminal

Step 4 — Clone the repo

```
git clone https://github.com/[partner]/[repo]
```

```
cd [repo]
```

Confirm these files and folders are present after cloning:

```
code/01_pull_data.py
```

```
code/02_clean.py
```

```
code/03_descriptives.py
```

```
code/04_regression.py
```

```
references/library.bib
```

```
references/apa.csl
```

```
research_note.qmd
```

```
Taskfile.yml
```

```
requirements.txt
```

```
.env.example
```

```
README.md
```

■ If `data/` or `research_note.pdf` appear in the repo, your partner has committed files they should not have. Note this — it is a gitignore error.

Terminal

Step 5 — Create and activate a virtual environment

```
python -m venv .venv
```

```
.venv\Scripts\activate
```

■ *Mac / Linux:*

```
python3 -m venv .venv
```

```
source .venv/bin/activate
```

Your prompt should show `(.venv)`. If not, the activation failed — check you are in the repo root.

Terminal

Step 6 — Install dependencies

```
pip install -r requirements.txt
```

This should complete without errors. Common issues:

```
wrds install fails -> pip install wrds --upgrade
```

```
linearmodels fails -> pip install linearmodels==5.4
```

```
any other failure -> note which package and tell your partner
```

■ Do not fix dependency errors yourself — note them and tell your partner. A broken `requirements.txt` is an issue they need to fix before submission.

Terminal

Step 7 — Register the Jupyter kernel

Run this once so Quarto can find the kernel when rendering:

```
python -m ipykernel install --user --name=sme-venv --display-name  
"SME Project (.venv)"
```

■ This only needs to be done once per machine. If your partner has named their kernel differently in the YAML, you may need to adjust.

VS Code

Step 8 — Create your .env file with your WRDS credentials

Copy the example file and add your own WRDS username:

```
copy .env.example .env
```

■ **Mac / Linux:**

```
cp .env.example .env
```

Open .env in VS Code and set your username:

🔗 **.env**

```
WRDS_USERNAME=your_wrds_username_here
```

■ **Use YOUR OWN WRDS credentials — not your partner's. The .env file must never be committed to Git.**

C Run the full pipeline

Terminal

Step 9 — Run task all

```
task all
```

This should run the complete chain: pull → clean → descriptives → regression → quarto render. Watch the terminal output and note any errors.

■ **This will take 10-30 minutes if it runs the full WRDS pull. If the pull step fails with a WRDS connection error, run each step individually (see Step 10).**

Terminal

Step 10 — If task all fails — run steps individually

If the pull fails (WRDS timeout is common), run each script manually and note at which step the failure occurs:

```
python code/01_pull_data.py
```

```
python code/02_clean.py
```

```
python code/03_descriptives.py
```

```
python code/04_regression.py
```

```
quarto render research_note.qmd --to typst
```

■ **A failure in 01_pull_data.py that says 'connection refused' or 'timeout' is a WRDS network issue, not a code error — it may work on retry. A Python error (traceback) is a code error — note it and tell your partner.**

VS Code

Terminal

Step 11 — Verify all output files were created

After the pipeline completes, check that all expected output files exist:

```
output/tables/summary_statistics.csv ✓ or ✗
```

```
output/tables/regression_results.csv ✓ or ✗
```

```
output/figures/correlation_matrix.png ✓ or ✗
```

```
output/figures/dv_distribution.png ✓ or ✗
```

```
output/figures/main_relationship.png ✓ or ✗
```

```
data/processed/panel_clean.parquet ✓ or ✗
```

```
data/processed/panel_with_vars.parquet ✓ or ✗
```

```
research_note.pdf ✓ or ✗
```

■ **Tick each one. Any missing file means a script failed silently — check the terminal output for warnings.**

D Open and verify research_note.pdf

VS Code

Step 12 — Open the PDF and check the title page

Open research_note.pdf and verify the title page contains:

Title: present and correct	↴
Subtitle: present	↴
Author name: present	↴
GitHub URL: visible on title page (not in introduction)	↴
Date: present	↴

■ If the GitHub URL appears in the introduction text instead of the title page, the author YAML field is not correctly formatted. Note this.

VS Code

Step 13 — Check section structure and formatting

Section numbers: 1, 2, 3, 4, 5 auto-generated ✓ or ✗	↴
All five sections: Introduction, Theory, Data, Results,	↴
Discussion/Conclusion present ✓ or ✗	↴
Citations: (Author, Year) format throughout ✓ or ✗	↴
NOT [@key] literals anywhere ✓ or ✗	↴
Equation: regression equation in LaTeX rendered ✓ or ✗	↴

VS Code

Step 14 — Check tables and figures

Summary statistics table: present and readable ✓ or ✗	↴
Regression table: all three models present ✓ or ✗	↴
Main relationship figure: visible, axis labels readable ✓ or ✗	↴
Correlation matrix: visible ✓ or ✗	↴
Inline numbers: actual values, not [N] placeholders ✓ or ✗	↴

■ If a table shows as raw code output rather than a formatted table, the Python chunk failed silently. Check the terminal output from task all for warnings.

VS Code

Step 15 — Check the results interpretation

Read the Results section and verify:

H1 explicitly stated as SUPPORTED or NOT SUPPORTED ✓ or ✗	↴
H2 explicitly stated as SUPPORTED or NOT SUPPORTED ✓ or ✗	↴
OLS vs FE difference discussed with % magnitude ✓ or ✗	↴
Every main coefficient has a plain-language interpretation ✓ or ✗	↴
References section present at the end ✓ or ✗	↴

E Report findings to your partner

VS Code

Step 16 — Fill in the summary below and share with your partner

Complete the table and send it to your partner so they can fix any issues before the deadline:

Check	Result	Notes
Repo structure correct (no data/ committed)	■ Pass ■ Fail	
pip install -r requirements.txt succeeded	■ Pass ■ Fail	

task all completed without errors	■ Pass ■ Fail
All output files generated	■ Pass ■ Fail
research_note.pdf produced	■ Pass ■ Fail
GitHub URL on title page (not introduction)	■ Pass ■ Fail
Citations formatted (Author, Year) not [@key]	■ Pass ■ Fail
Tables and figures render correctly	■ Pass ■ Fail
H1 and H2 stated as supported / not supported	■ Pass ■ Fail
No [N] or placeholder text remaining in PDF	■ Pass ■ Fail

■ Terminal

Step 17 — Clean up after the peer check

Delete the cloned repo when you are done — do not leave someone else's data on your machine:

```
cd ..
```

```
rmdir /s /q [repo-name]
```

■ *Mac / Linux:*

```
cd .. && rm -rf [repo-name]
```

■ *Also delete the .venv folder inside the cloned repo before removal to avoid permission errors on Windows.*

Why this matters: The examiner will follow exactly these steps on your repo. A pass on every check here means your submission is fully reproducible. Any fail is something you can fix before June 28 — but not after. Deadline: June 28, 2026 — 23:59. Reference: github.com/vkiefner/sme-intl